

Ans to the Q: NO: 1

Role of Sensors and Actuators in Robotic system.

Sensors and actuators are two essential part of a Robotic system.

Sensor: A sensor is an input device that collect information from the robot's environment. It converts a physical condition, such as distance, temperature, sound, light, or moisture, into an electrical signal that a microcontroller can process.

examples:

- ① Ultrasonic Sensor: Distance from a object. It's uses for the obstacle detection.
- ② Soil Moisture Sensor: Amount of water in soil. uses for Smart irrigation.
- ③ Temperature Sensor: Environmental temperature. Greenhouse monitoring.
- ④ Humidity Sensor: Moisture in Air. Agriculture monitoring.
- ⑤ IR Sensor: Infrared light or nearby objects. Line following.
- ⑥ Microphone Sensor: Sound, Voice or Sound detection.
- ⑦ Camera: Images and video. Object Recognition.

These Sensor are used in a Robot and there are also other sensors which is used a Robot and help a Robot to understand the Nature.

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Actuators: An actuator is an output device that creates physical movement or action according to a command from a microcontroller.

- ① DC motor: Rotational movement. Moving Robot wheels.
- ② Servo: Controlled angular movement. Moving Robotic Arm.
- ③ Water pump: Pumps water. Automatic irrigation.
- ④ Relay: Switches high-power devices. Controlling a pump.
- ⑤ Buzzer: produces sound. Warning or notification.
- ⑥ LED: produces light. status indication.

Interaction with Microcontroller :-

The microcontroller acts as the brain of the Robot.

⇒ Environment → Sensor → Microcontroller → Decision
↓
Action ← Actuator

Think about the Krishi-Rover:

- ① Soil moisture sensor measure the moisture level.
- ② Sensor send the value to the ESP32.
- ③ ESP32 compares the value with a predefined threshold.
- ④ if the soil Dry, ESP32 activates the Relay.
- ⑤ the Relay turn on the water pump.
- ⑥ When enough moisture is Detected, the pump is turned off.

The Sensor allow a Robot to "perceive", the microcontroller allows to "Think" and the Actuators allow to "act".

Sensor = Input ; Microcontroller = Brain ; Actuator = Output.

Ans to the Q! NO! 2

Autonomous Robot: An autonomous robot can sense its environment, make decisions, and perform tasks without continuous human control.

⇒ Robot vacuum cleaner, Self driving agriculture Rover, Warehouse delivery Robot, Automatic irrigation Robot, Autonomous Robot.

If we think about the Krishi-Rover it automatically avoid obstacle, changes its direction, checks soil moisture and activates irrigation.

Advantage of Autonomous Robot:

- ① Require less human effort.
- ② Can work continuously.
- ③ produces consistent results.
- ④ useful in Dangerous environments.
- ⑤ Can response quickly to sensor data.

Disadvantage:

- ① More expensive and complicated.
- ② Sensor failure may cause incorrect decisions.
- ③ Difficult to handle completely new situations.
- ④ Require more testing.
- ⑤ Raises safety and accountability concerns.

(4)

Semi Autonomus Robot : A semi-autonomus robot can perform some task independently, but a human operator is still required for supervision, task selection, approval, or manual control.

Example:

A drone controlled by a pilot but capable of maintaining attitude automatically.

Surgical Robot Controlled by a Surgeon.

Advantage:

① Human can intervene during emergency

② Easier to develop

③ More suitable for changing environments

④ usually less expensive

Disadvantage:

① Required an operator

② Human error can occur

③ Cannot work completely independently

④ Communication Delay.

Comparison:

Autonomus Robot	Semi-autonomus Robot
Human involvement very Low	Human involvement Very High
Decision making mostly by Robot	Decision making shared by Robot and human.
Flexibility Depends on Programmed intelligence	flexibility Depends on Human can manage unexpected situations.
Safety Requires Reliable Sensor and algorithm	Safety human can intervene
Cost usually Higher	Cost usually lower

Ans to the Q! NO! 3

Wireless communication method in Robotics:

It's allow a Robot communication - to exchange commands, sensor reading, images, and status information without physical cables.

① Bluetooth: that's commonly used for short Range Communication between a robot and a smartphone, laptop or controller.

Application:

- ① Manual Robot Control ② Sending simple document
- ③ Reading sensor data ④ changing Robot operation mode.

② Wifi: that is used when a robot needs internet access, dashboard communication, cloud storage, video transmission, or control over a local network.

Application:

- ① Web based Robot Control ② IoT monitoring ③ Cloud Databases
- ④ Live sensor dashboard ⑤ Software updates.

③ Radio frequency (RF): that modules such as 433 MHz, nRF24LD1, LoRa, and Zigbee are used for long range or low power communication.

Application:

- ① Agriculture fields ② Drone Control ③ Remote environmental.
- ④ Communication in areas without Wi-Fi.

④ Cellular Communication: GSM, 4G or 5G can be used when a robot operates outside the range of Wifi.

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Bluetooth and Wi-Fi Comparison :-

Feature	Bluetooth	Wi-Fi
Typical Range	Short to medium	Medium
Data speed	Low to moderate	High
Power Consumption	Lower	Higher
Internet Connection	Normally not Required	Can provide internet Access
Setup	Simple device to device pairing.	Require network or access point.
Best use	Manual Control and Small Sensor Data	Dashboard, cloud and large data.
Robot Example	Smartphone-Controlled Rover	IoT based Rover.

Bluetooth that's depends on the Range Radio, power, antenna, physical layer, interference and obstacle. That's designed for the low energy and low power operations.

Wifi is designed for Higher-throughput Communication but consumes more power. ESP32 conveniently integrates both Wi-Fi and Bluetooth, making it suitable for service robot and smart agriculture application.

Ans to the Q! NO! 4

Dec model of emotion invented by Andrew Ortony, Gerald Clore and Allan Collins.

Dec = Ortony Clore Collins

It's a Cognitive appraisal model of emotion. This means that emotion is generated after a person evaluates or interprets an event, action, or object.

The model doesn't simply ask "what happened?" It asks:

- * Is the event good or bad for my goal?
- * Is a person's action praiseworthy or blameworthy?
- * Do I like or dislike the object?

Three Main Concepts:

① Consequences of Events:- These emotions occur when a person evaluates the outcome of an event.

Joy, Distress, Hope, Fear, Relief, Disappointment.

As an example A farmer feels joy when the Rover successfully waters the dry plants. The farmer may feel distress if the robot damages the crop.

② Actions of Agents:- These emotions depend on whether someone's action is considered praiseworthy or blameworthy.

Pride, Shame, Admiration, Reproach.

A developer may feel pride after designing a successful obstacle-avoidance system.

③

① Aspects of Objects: These emotions are based on liking or disliking an object.

Love or liking, Hate or disliking.

A farmer may like Knishi-Rover because it is useful easy to operate.

Oce Process:

Event, Action or Object → Cognitive Evaluation
↓
Emotion

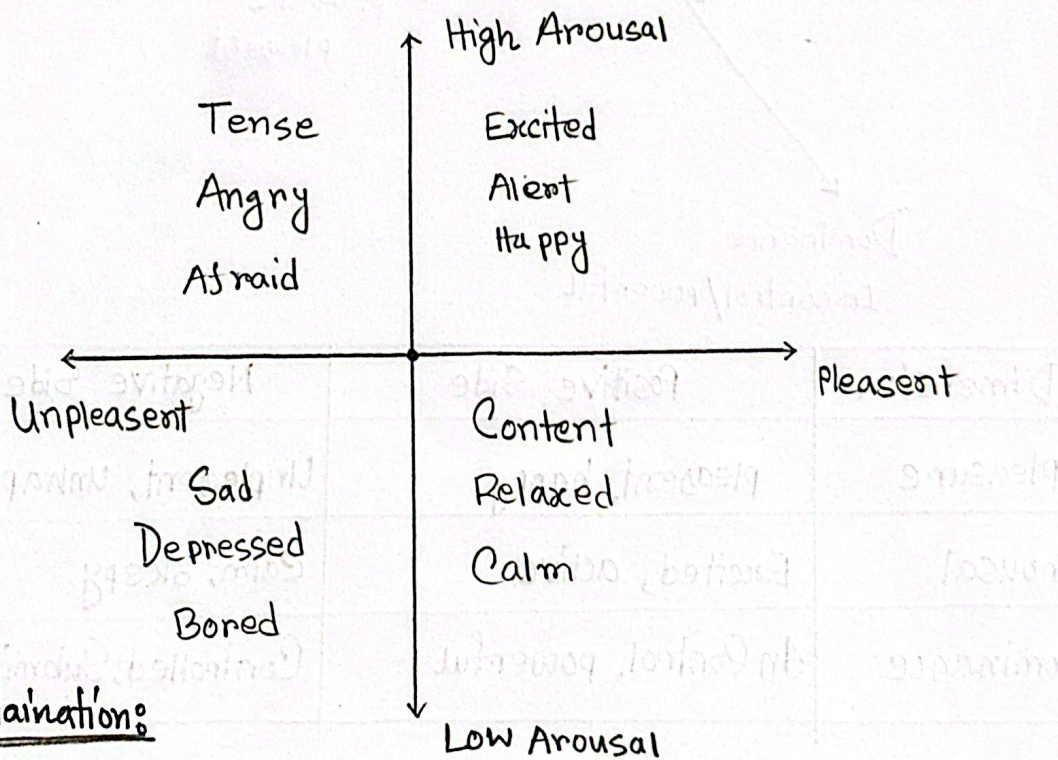
Oce emotion depends on how a person evaluates events, actions and objects.

Emotion Models:-

a) Russell's Circumplex Model of Affect Represents emotions using two dimensions.

① Valence :- pleasant to Unpleasant.

② Arousal :- High activation to Low activation.



Explanation:

Happy → pleasant and highly active.

Excited → Very pleasant and highly aroused.

Relaxed → pleasant but low arousal.

Sad → Unpleasant Low Arousal

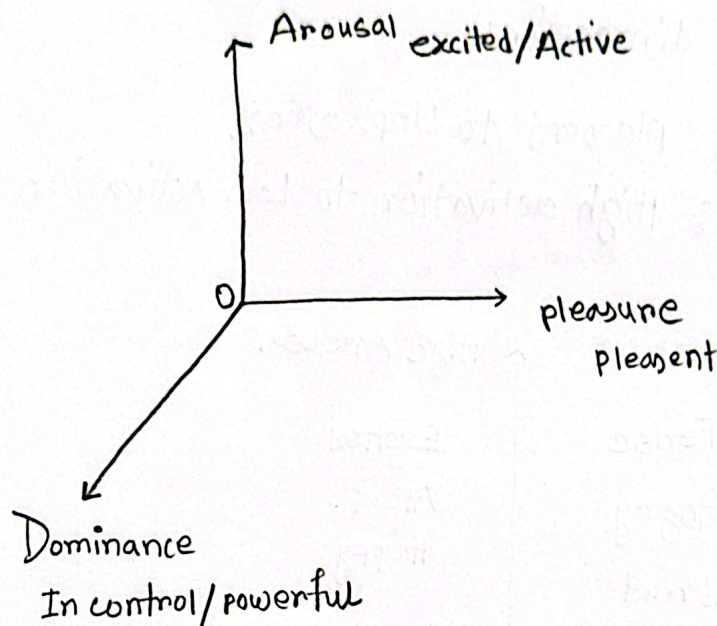
HRI: If a Robot moves suddenly towards a user, the user may experience fear, which has negative valence and high Arousal.

If the Robot speaks slowly and provides useful assistance, the user may feel calm or satisfied, which has positive valence and low arousal.

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b) PAD emotion Model: Pleasure, Arousal, Dominance.

It was developed by Albert Mehrabian and James Russell as a dimensional framework for emotional states.



Dimension	Positive Side	Negative Side
Pleasure	pleasant, happy	Unpleasant, unhappy.
Arousal	Excited, active	Calm, sleepy
Dominance	In Control, powerful	Controlled, Submissive

Difference between Russell's Model and PAD

Russell's Model	PAD model
uses two dimensions	uses two dimensions
Valence and arousal	pleasure, arousal and dominance
Easier to draw	Describes control or power as well.
Emotion appear in a Circle	Emotions appear in three-dimensional space.
Russell = Valence + Arousal	PAD = pleasure + Arousal + Dominance.

Ans to the Q! NO! 6

① Qualitative Method :- A qualitative method collects non-numerical information to understand people's opinions, experiences, feelings, behavior or reasons.

Data collection methods :-

- ① Interviews ② Focus-group discussion ③ Observation
- ④ Open-ended question ⑤ participant feedback.

Advantage :-

- ① provide detailed explanation.
- ② Helps researchers understand user experiences.
- ③ useful for identifying unexpected problems.

Limitations :-

- ① usually involve fewer participant.
- ② Analysis maybe time consuming.
- ③ Result maybe influence by personal interpretation.

② Quantitative Method :- A quantitative collects numerical data that can be measured and statistically analysed.

Data Collection Method :-

- ① Structured Surveys ② Sensor Measurements ③ Experiments.
- ④ Task-Completion time ⑤ Rating scale.

Advantage :-

- ① Results are measurable.
- ② Easy to compare.
- ③ Can include many participants.

Limitation:

- ① May not explain why a problem occurred.
- ② Important user feelings may be missed.
- ③ poor Result may be wrong sometimes.

① Mixed Method:- A mixed Method combines qualitative and quantitative methods in the same study.

In one Research do both Qualitative and Quantitative very useful cause it helps a Researcher integrates numerical and Non-Numerical evidence.

Comparison:

Method	Type of Data	Example
Qualitative	Words, Opinions, experiences	Farmer interview.
Quantitative	Numbers and Measurements	Sensor accuracy.
Mixed	Both words and Numbers	Survey + interview.

For our project both are useful and we have done it to make it Futuristic and Realistic implicable Project.

Ans to the Q. NO. 7Ethical Consideration of HRI Studies:

HRI means Human-Robot Interaction. Ethical consideration protect participants and ensure that the robot is developed and responsibly.

① Informed Consent: Participant must be understand

- ① The purpose of the Research.
- ② What they will be asked to do.
- ③ possible Risk.
- ④ What data will be collected.
- ⑤ Their right to stop participating.

② Privacy: A Robot may collect voice Recordings, images, location, behaviour, or personal information. Researchers must not collect unnecessary data.③ Data Security: collected data should be →

- ① password-protected.
- ② Stored securely.
- ③ Shared only with authorize people.
- ④ Deleted when no longer required.

④ physical Safety:

- ① Emergency stop button.
- ② Low operating speed near human.
- ③ properly covered moving parts.
- ④ Automatic shutdown during failure.

⑤ Psychological Safety: The robot should not create unnecessary fear, stress, pressure, or emotional manipulation. Researchers should explain that the robot is a machine, not a human being.

⑥ Transparency: user should know

- ① whether the robot is autonomous.
- ② what data it collects.
- ③ what its limitations.
- ④ how decisions are made.

⑦ Avoiding Deception: Researchers should not claim that a robot has human level feelings, emotions, understand.

⑧ Fairness and Bias: The Robot should be tested with different groups of user.

⑨ Right to Withdraw: Participants must be free to leave the study at any time without punishment.

⑩ Responsibility and Accountability: The Research team must be clearly determine who is responsible if the Robot:

- ① causes injury.
- ② Damage Crops.
- ③ records private information.

Ans to the Q: No: 8

Two sectors where Robotics can create significant socio-economic impact are:

- ① Agriculture
- ② Ready-made garments and manufacturing.

Agriculture: Agriculture supports a large scale section of Bangladesh workforce and rural economy. Farmers commonly face labour shortages, high labour costs, climate related uncertainty, inefficient irrigation, difficulty monitoring large fields. World Bank research has highlighted both the continuing employment importance of agriculture and the role of mechanization and agri-technology in improving productivity.

Possible Robotics Application:

- ① Soil monitoring
- ② Automatic irrigation
- ③ Seed planting.
- ④ Fertilizer application
- ⑤ Weed detection
- ⑥ fruit harvesting.

Socio economic Benefits:

- ① Reduces physically demanding work
- ② Saves water.
- ③ Improves crop monitoring
- ④ Reduces production costs.
- ⑤ Supports precision farming.

Ready-made Garments and Manufacturing:

Bangladesh's RMG sector is its main export industry and important source of formal employment. Automation is already changing production, which makes workers reskilling and responsible technology adoption essential.

Possible Robotic Application:

- (i) fabric cutting (ii) Material handling (iii) Quality inspection
- (iv) packaging (v) Warehouse transporting (vi) fire and Safety.
- (vii) Dangerous chemical handling.

Socio Economic Benefit:

- (i) improves product quality (ii) increases production speed.
- (iii) Reduces Material waste (iv) protect workers from dangerous tasks.
- (v) Create technical jobs in Robot operation and maintenance.

In Conclusion, Automation should not only replace workers. factories and government organization should provide training in

- (i) Robot operation,
- (ii) maintenance.
- (iii) Quality Control.
- (iv) programming.
- (v) Industrial Safety.

Ans to the Q: NO: 9

Our project is Krishi-Rover: An autonomous Farming Rover for soil monitoring, Smart irrigation, Bangla voice Control.

I am Akhi Ketu Chakma. and I am the full stack Contributor of the project. I covered everything and gave my team the ideas about the project.

A) Literature Review: For the Review I studied the previous papers.

- ① Agriculture Robot ② obstacle detection system.
- ③ IoT based irrigation ④ Esp 32 working Demo
- ⑤ How to integrate Bangla voice and Algorithms.

It Helps me to find out the whole project context how we can build this and the Gap between existing and our system.

B) Hardware System: For the Hardware implemented the Rover from scratch. Working on selecting, connecting and testing each components.

- ① ESP 32 ② Four gear DC motors ③ L298N motor Driver
- ④ Buck Converter ⑤ Ultrasonic Sensor ⑥ Relay module.
- ⑦ Water pump ⑧ Speaker ⑨ Amplifier module
- ⑩ Humidity sensor ⑪ Capacitive soil moisture sensor.
- ⑫ Battery pack ⑬ LED ⑭ Microphone. Etc.

(C) Software System : For the software first i built the whole website using Typescript, React, python, Js. Hosted or Deployed in Vercel. Another part ESP32 and High Level Bangla Speech.

Embedded Software: ESP32 Performs:

- ① Sensor Reading ② Distance Calculation ③ Motor Control
- ④ Obstacle avoidance ⑤ pump Control ⑥ status reporting.
- ⑦ Wi Fi or Bluetooth communication.

High Level Software:

- ① Microphone input ② Bangla STT ③ Command identification
- ④ Command transmission ⑤ BanglaTTS ⑥ Sensor data.

(D) Image and Table Generation:

- ① Hardware Diagrams ② Wiring Diagrams ③ Flowcharts
- ④ Algorithms ⑤ Component tables ⑥ presentation figures.

(E) Algorithms:

- ① obstacle avoidance ② Soil-moisture Irrigation
- ③ Voice Command Algorithm ④ priority based decision.

(F) Results: During the prototype showing to Sir, that's able to do the desired things. and There's little update we have to add the AI-agent must be Real time talking speech.

Ans to the Q: NO: 10

Classification of the ten Most Relevant works.

Our project is Krishi-Rover:

Old works	How it connects
An overview of smart irrigation systems using IoT (2022).	Supports the IoT-based irrigation concept by explaining smart irrigation architecture, sensing and water saving methods.
IoT-Enabled Smart Agriculture: Architecture, Application and Challenges. (2022).	Connects with the ESP-32 based IoT monitoring system.
Evaluation of IoT Based Smart Drip irrigation and ETC based System for Sweet corn (2023).	Support efficient irrigation decision-making and helps justify watering only when needed.
IoT-Enabled Smart Agriculture for improving water management (2025).	Related to water management in agriculture and supports the adaptive spot watering system.
Visual Navigation and obstacle Avoidance Control for Agriculture Robots (2023).	Connects with autonomous agriculture robot movement and obstacle avoidance for rover navigation.
Comparative Analysis of Soil-moisture -and weather-based irrigation scheduling (2025)	Supports soil moisture-based irrigation scheduling, directly matching the moisture sensor based watering logic
A Comprehensive Review of Obstacle Avoidance for Autonomous Agricultural Machinery. (2025)	Supports autonomous navigation and obstacle detection research background.

Old Works	How it Connects
Low Cost AI-Driven Autonomous Rescue Bot for water-based Life saving Mission (2025)	Connects with low-costs autonomous robot development and decision-making logic.
IoT based fruit quality inspection and Lifespan Detection system (2024)	Supports IoT based agricultural monitoring and sensor-based decision making.
Smart floor cleaning Robot (2023)	Connects with mobile Robotics, motor control, obstacle handling, and autonomous movement.

for my project "Krishi-Rover" that's a prototype right now. So, if we build with the high Accessories it might be useful in Bangladesh Real life based.

- ① Agricultural Farms → Detect Dry Zone and Update immediate
- ② Plant Nurseries → That Small a movement friendly might be perfect.
- ③ Greenhouses → Temperature, Humidity and Soil Condition
- ④ Rooftop Gardens
- ⑤ Voice implement easy to control and user friendly.

So, overall that's basically designed for the future project and must be helpful for the farmers.

Ans to the Q: NO: 11Data Processing, Decision making and Integration Challenges:

Krishi-Rover Collects the data from Ultrasonic, soil-moisture, temperature and humidity sensors. The ESP 32 processes these readings and compares them with predefined conditions.

Data collection → Data Conversion → Data filtering

↓

Feedback ← Actuation ← Decisions ← Comparison Threshold

Decision Making Technique :

- ① Rule-based → if, else if, else
- ② Threshold-based → If soil moisture < dry threshold :
pump ON.
- ③ Priority-based → Safety decision always first.
- ④ Finite-state → Different states : Moving Forward, Turning Left etc.

Hardware - Software Integration Challenges :

- ① Voltage difference : ESP32 GPIO pins normally operates 3.3V, while some sensor may provide a 5V signal. A voltage divider shifted
- ② Motor Speed : Motor rotation and speed may be Rotate different direction.
So, use the IDE fix the issue. Calibrate the motors.
Adjust the PWM values.

- ③ ultrasonic Interference: There is three ultrasonic sensor so, sometimes may be it's delayed detection and showing IoT value late.
fixing this using the software update code.
- ④ Power Supply: Motor pumps need more current than esp32 can supply.
fixing this using Relay, motor driver, suitable power source.
- ⑤ Voice Recognition Noise: Agriculture environments contain motor noise, wind, human voice.
- ⑥ Communication Delay: With disconnection may delay the remote commands.
- ⑦ Mechanical Problem: Sensor may moves during the vibration.

So, Overall the project is working perfectly with the all equipments. We expected that if we invest on it to make real field types it must be perfect and efficiently work for farmers.

Ans to the Q! NO! 12

Knishi-Rover Interacts with the environment through a continuous loop.

Sense → Process → Decide → Act → Sense → ... →

- ① Sense: The robot measures → Obstacle distance, Soil moisture temperature, Humidity,
- ② Processing: Converts sensor signal
- ③ Decision: Move forward, left, Right, backward, irrigation.
- ④ Action: The motor moves the rover, while the relay and pump perform irrigation.
- ⑤ Feedback: The Robot sends its condition to user through.

Key Algorithm:

- ① Obstacle avoidance.
- ② Automatic irrigation.
- ③ Hysteresis Control.
- ④ Finite state Machine.
- ⑤ Voice Command Matching.

Programming Language:

- * C or C++ using Arduino framework.
- * Python used for Bangla voice Assistant and voice interaction High level software.
- * Javascript or Typescript, these things are basically used for the web dashboard.

Libraries :

- Arduino.h → basic esp32 Function.
- wifi.h → WiFi Communication.
- Bluetoothserial.h → Bluetooth Communication.
- Webserver.h → Local web server.
- DHT.h → Temperature and Humidity.
- Newping.h → Ultrasonic - sensor control.
- Speech.recognition → Capturing and recognition speech.
- gTTS → Bangla Text-to-speech.
- pygame → playing generated audio.
- requests → Sending HTTP commands.
- pyserial → Serial communication.
- Flask or FastAPI → Creating a local API.

These things are chosen cause everything are Related to our project and I wrote everything how it connects to our project.

- * ESP32 is affordable and contains wifi and Bluetooth.
- * c/c++ provides direct and fast control of hardware.
- * python simplifies Bangla voice processing.
- * Web interface for IoT and communication.

_____The End_____